



BATCH : **B146-149 CS & DevOps**
LESSON : **Network**
DATE : **07.07.2023**
SUBJECT : **IP Subnetting**

ZOOM GİRİŞLERİNİZİ LÜTFEN **LMS** SİSTEMİ ÜZERİNDEN YAPINIZ



Previous Session

- IP Address Blocks
- Network ID/Address
- Host ID/Address
- Subnet ID/Address
- Subnet Mask
- AND operation
- NAT



Contents

- IP Subnetting
- CIDR

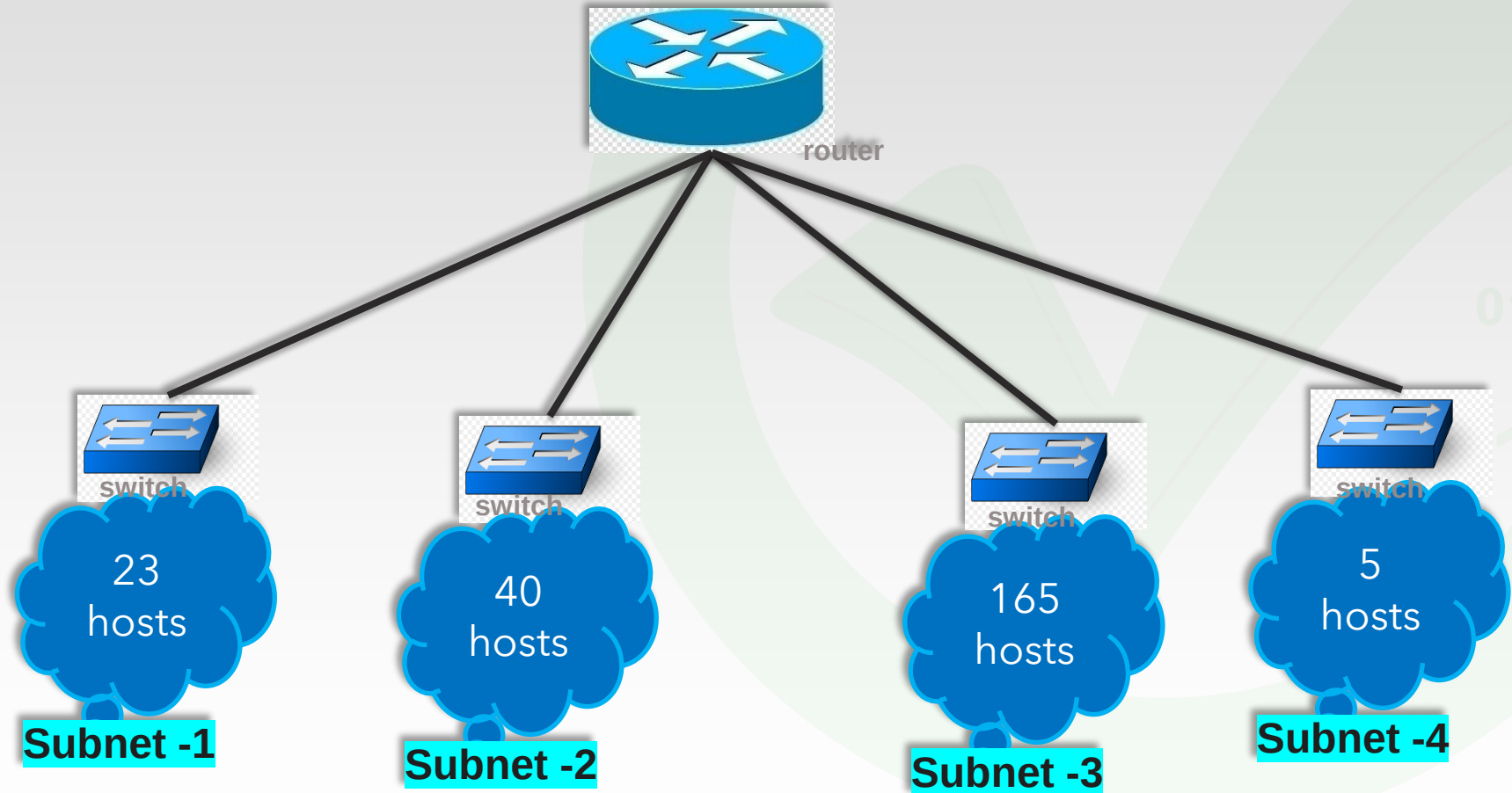


NETWORK Day 6



What is IP Subnetting?

- ❓ A subnetwork or **subnet** is a logical subdivision of an **IP** network. The practice of dividing a network into two or more networks is called **subnetting**.





Why IP Subnetting?

- **Subnetting** helps to reduce the network traffic and conceals network complexity. This way, traffic doesn't have to flow through unnecessary routes, increasing network speeds.
- **Subnetting** is essential when a single network number has to be allocated over numerous segments of a local area network (LAN).
- **Subnets** were initially designed for solving the shortage of IP addresses over the Internet.
- **Routers** are used to communicate between subnets.
- **Routers** use **subnet IDs** to route traffic.

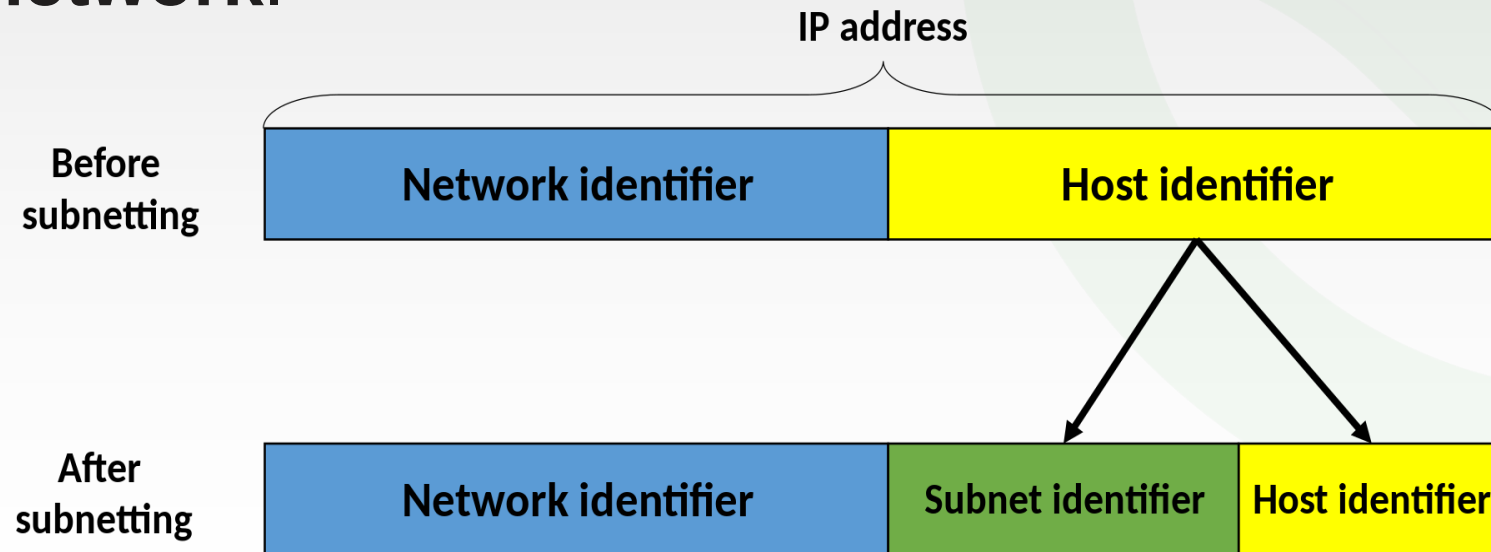


- [illegible]



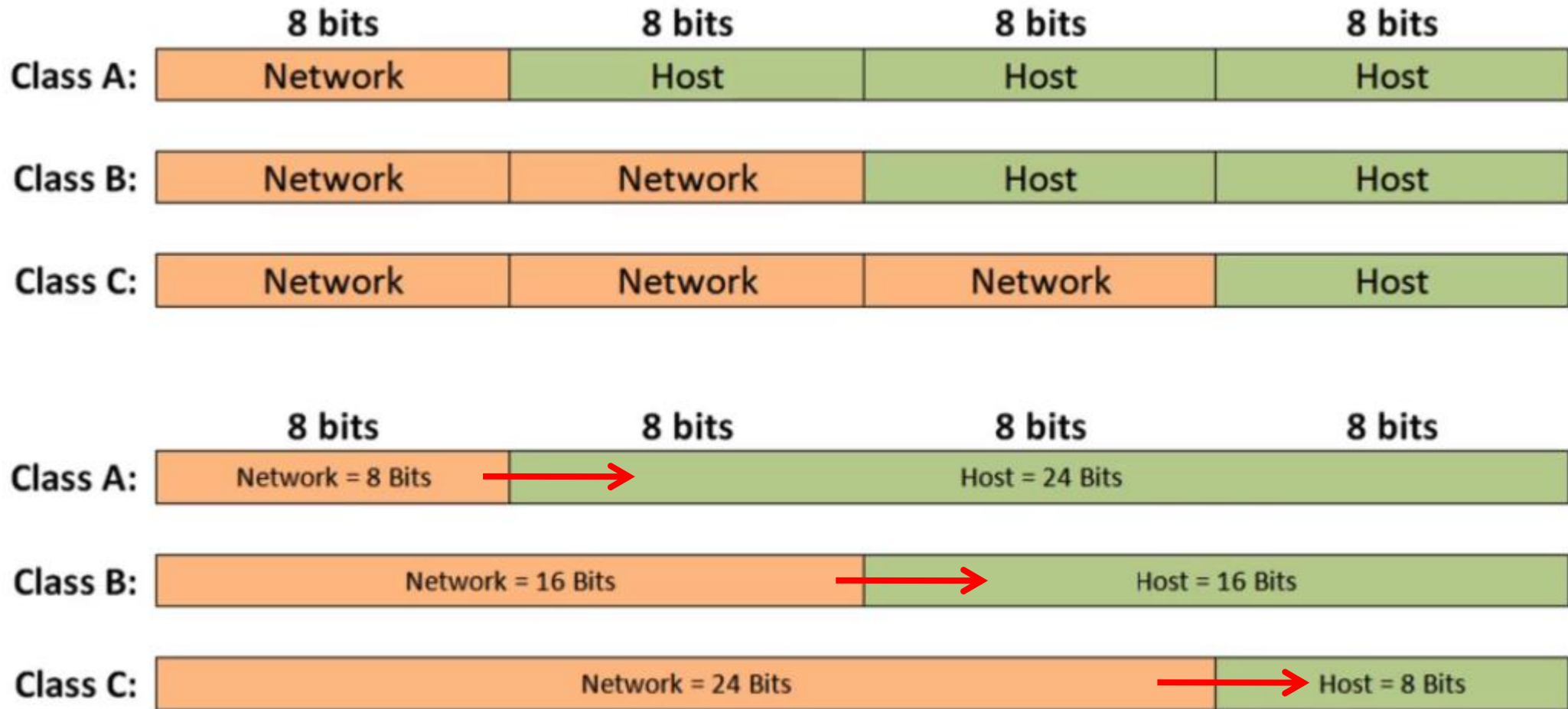
How does it work?

- IP subnetting results in the logical division of an **IP** address into two fields: **the network number** or routing prefix and the rest field or **host identifier**.
- In subnetting we **borrow some bits from host-identifier** to use as **subnetwork**.





Subnetting IPv4 Addresses





-
- The diagram illustrates the binary representation of the IP address 192.168.1.1 and the subnet mask 255.255.255.192. The IP address is shown as 1 1 0 0 0 0 0 0 | 1 0 0 0 0 0 0 0 | 0 0 0 0 0 0 0 0 | 0 0 | 0 0 0 0 0 1. The subnet mask is shown as 1 1 1 1 1 1 1 1 | 1 1 1 1 1 1 1 1 | 1 1 1 1 1 1 1 1 | 1 1 | 0 0 0 0 0 0. Brackets below the binary strings indicate the Network part (the first three octets), the Subnet (the fourth octet), and the Host part (the last octet).
- | IP address | 192.168.1.1 | | | | | | |
|------------|-------------|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
- | Subnet mask | 255.255.255.192 | | | | | | |
|-------------|-----------------|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
- Network part Subnet Host part



Subnetting IPv4 Addresses

Binary (N.N.N.H)	Decimal	CIDR	# Subnets (2^x)	Block Size (2^y)	# Hosts ($2^y - 2$)
N.N.N.00000000	255.255.255.0	/24	$2^0 = 1$	$2^8 = 256$	$2^8 - 2 = 254$
N.N.N.10000000	255.255.255.128	/25	$2^1 = 2$	$2^7 = 128$	$2^7 - 2 = 126$
N.N.N.11000000	255.255.255.192	/26	$2^2 = 4$	$2^6 = 64$	$2^6 - 2 = 62$
N.N.N.11100000	255.255.255.224	/27	$2^3 = 8$	$2^5 = 32$	$2^5 - 2 = 30$
N.N.N.11110000	255.255.255.240	/28	$2^4 = 16$	$2^4 = 16$	$2^4 - 2 = 14$
N.N.N.11111000	255.255.255.248	/29	$2^5 = 32$	$2^3 = 8$	$2^3 - 2 = 6$
N.N.N.11111100	255.255.255.252	/30	$2^6 = 64$	$2^2 = 4$	$2^2 - 2 = 2$

Number of Subnets (2^x)

- X = number of host bits we borrow to create subnets

Block Size (2^y)

- Y = number of remaining host bits left that are used for the subnet IP addresses

Hosts per Subnet ($2^y - 2$)

- There are two addresses per network (or subnet) that we cannot use to assign to hosts on that network:
 - **Network Address:** This is the address used to uniquely identify the network (or subnet).
 - **Broadcast Address:** Address reserved for broadcast communication on the network.



Subnetting IPv4 Addresses

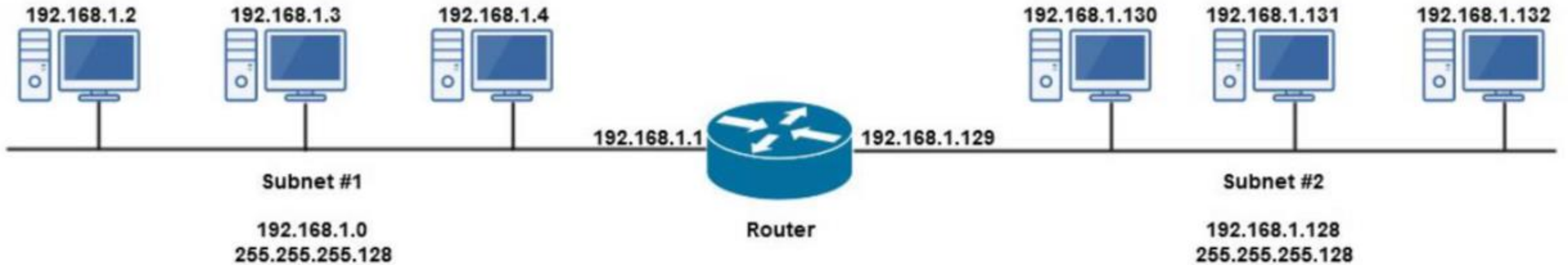
- **Details & Requirements**
 - Network Address: 192.168.1.0
 - Default Subnet Mask: 255.255.255.0
 - Requires 2 Subnets
- **How many host bit do we need to borrow?**
 - 1 host bit, $2^1 = 2$ Subnets
- **How many addresses hosts per subnet?**
 - 7 host bits left, $2^7 = 128$ Addresses / Subnet
 - $2^7 - 2 = 126$ Addresses / Subnet
- **What are the valid subnets?**
 - 192.168.1.0 and 192.168.1.128
- **New Subnet Mask?**
 - 11111111.11111111.11111111.10000000
 - 255.255.255.128 or /25

Subnet	#1	#2
Network Address	192.168.1.	192.168.1.
First Host IP	192.168.1.	192.168.1.
Last Host IP	192.168.1.	192.168.1.
Broadcast Address	192.168.1.	192.168.1.

See Subnet Calculator



Subnetting IPv4 Addresses





Subnetting IPv4 Addresses

192.168.1.1 and Mask 255.255.255.192

Broadcast address

➡ **192.168.1.63**

A Host/ip address

➡ **192.168.1.62**

Network address/id

➡ **192.168.1.0**



CIDR Classless Inter Domain Routing

IP address **192.168.1.0** and Mask **255.255.255.0**



This IP address is in CIDR notation, which is a compact way of including the subnet mask along with the address. The /24 tells you that the first 24 bits of the IP address are used for network routing.



CIDR



192.168.1.0 / 24

Total number of Hosts: $2^{32-24}-2 = 254$



CIDR

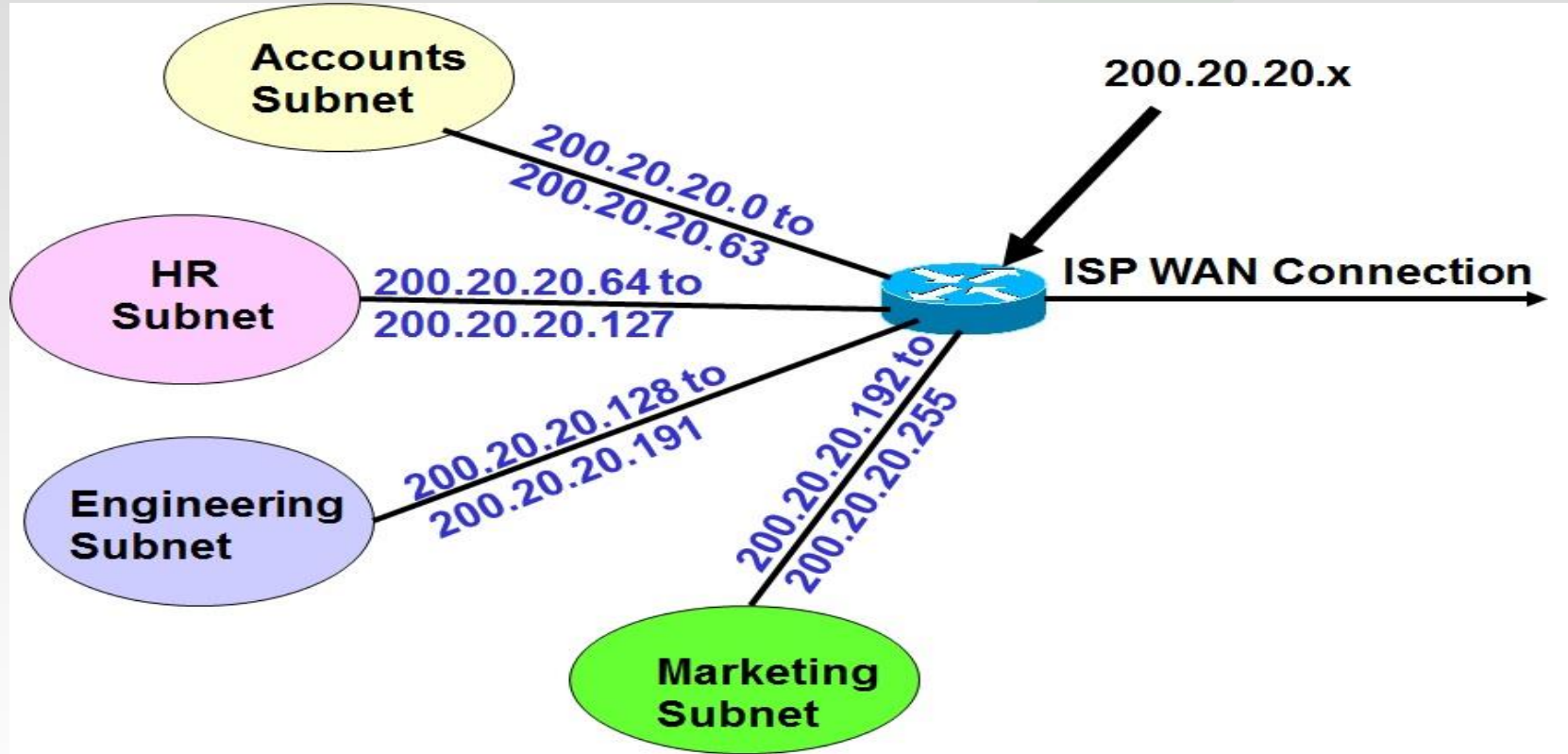


192.168.1.0 / 23

Total number of Hosts: $2^{32-23}-2 = 510$



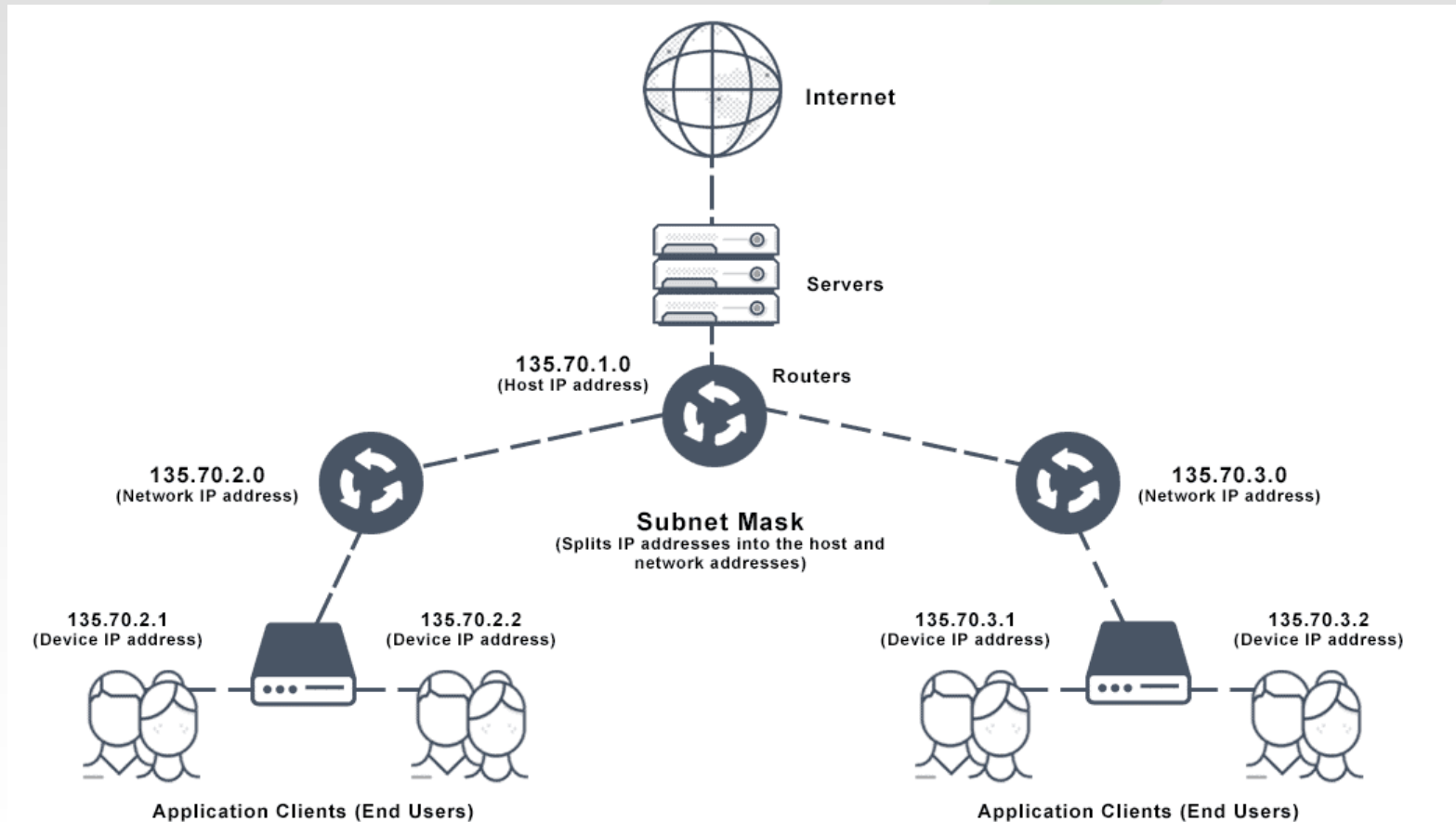
Examples



*A corporate network which has 4 subnets



Examples





Examples

Private IP address space	
From	To
10.0.0.0	10.255.255.255
172.16.0.0	172.31.255.255
192.168.0.0	192.168.255.255

10.0.0.1
10.0.0.2

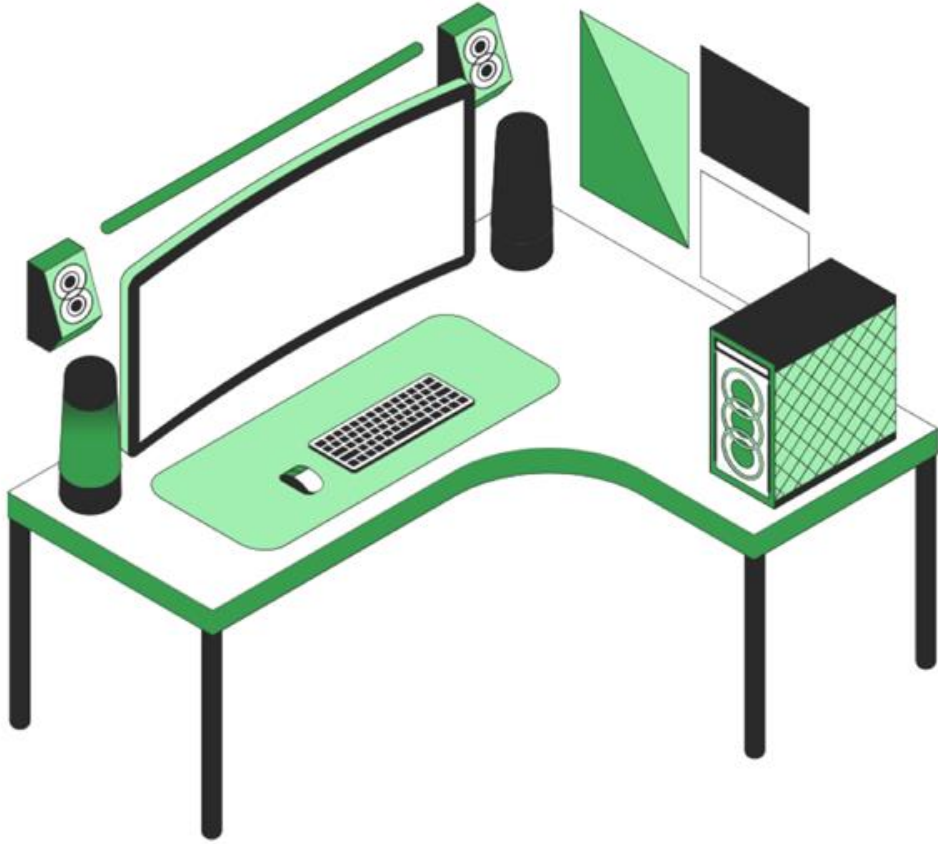
10.0.1.1
10.0.1.2

10.10.10.1
10.10.10.2

192.168.1.1
192.168.1.2

192.168.10.10
192.168.10.11

192.168.11.10
192.168.11.11



Do you
have any
questions?

Send it to us! We hope you learned something new.